

Agent-Based Modeling of Sexual Selection Dynamics in Evolutionary Systems

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Abstract

Sexual selection is a fundamental evolutionary mechanism that shapes trait development through mate choice, intra-sexual competition, and reproductive success. This paper investigates the role of sexual selection as a primary driver for sympatric speciation, specifically focusing on how localized reproductive decisions influence population-wide diversity. This paper analyzes the mechanisms of driving trait evolution and formulate key research questions regarding the dynamics of these traits under selective pressures. By examining the interplay between natural and sexual selection, this study outlines the theoretical frameworks necessary to model the spontaneous formation of evolutionary niches within complex, multi-modal fitness landscapes.

Keywords: sexual selection, trait evolution, sympatric speciation, mate choice, evolutionary biology, population dynamics.

1 Introduction

Evolutionary computation is a powerful paradigm for solving complex optimization problems by mimicking biological processes. However, classical Evolutionary Algorithms (EAs) often suffer from premature convergence and a loss of population diversity [1]. This limitation prevents the effective exploration of the search space and adaptation to dynamic environments, a phenomenon extensively documented in multimodal optimization studies.

To address these challenges, researchers turn toward the biological mechanisms of sympatric speciation and sexual selection. While natural selection focuses on the survival of the individual, sexual selection promotes traits that increase reproductive success, often leading to evolutionary innovations. This mechanism arises from sexual conflict and the asymmetry of reproductive costs typically higher for females, which necessitates the development of specific mate choice strategies. According to the theories of Charles Darwin [2], this process can be divided into two primary drivers:

- **Inter-sexual selection (Mate Choice):** involving the selection of individuals (usually by females) based on specific phenotypic traits perceived as indicators of high fitness or genetic quality.
- **Intra-sexual selection (Competition):** involving rivalry between individuals of the same sex for access to mates, leading to the evolution of competitive weaponry or displays.

In the context of Agent-Based Modeling (ABM), these biological principles offer a decentralized mechanism for niching and sympatric speciation. By simulating an environment where agents make autonomous mate choice decisions based on local information, it is possible to create a system where sub-populations naturally diverge [3]. This allows for the maintenance of so-called "useful diversity," which is central to models such as the Co-evolutionary Multi-Agent System (CoEMAS) and its formal extension, the Biological and Social Multi-Agent System (BSMAS) [4].

This project aims to implement and analyze such a multi-agent environment. The primary focus is on investigating how localized mating rules and competitive behaviors influence the long-term trajectory of trait evolution. The work serves as a tool for both biological simulation and the development of diversity-maintaining techniques in computational systems.

2 Previous Research on Sexual Selection as a Speciation Mechanism

Modern computational approaches to sexual selection have moved toward complex Agent-Based Modeling and specialized evolutionary frameworks. A significant advancement is the CoEMAS model, which treats sexual selection as a decentralized mechanism for maintaining diversity without the need for global control [4].

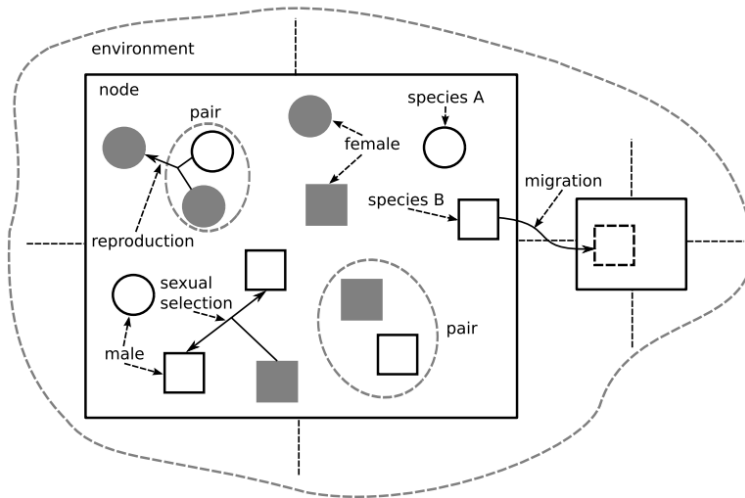


Figure 1: Multi-Agent System with Sexual Selection (source: [5])

Foundational theoretical models, such as those by Gavrillets, identify three general dynamic regimes: co-evolutionary chases, cluster splitting, and full reproductive isolation leading to the emergence of new species [3]. Furthermore, Todd and Miller suggest that sexual selection allows species to create their own optima in fitness landscapes, facilitating an "escape" from local optima and driving evolutionary innovations more efficiently than natural selection alone [6].

Recent implementations often utilize high-level simulation frameworks like Mesa to track thousands of agents and analyze population stability [7]. These models frequently incorporate signaling theory, specifically the Handicap Principle, demonstrating that the cost of a signal ensures its reliability. However, simulations also identify conditions under which dishonest

signaling (cheating) can become an evolutionary stable strategy, especially when verification costs are high.

The field is currently exploring Generative Agent-Based Modeling (GABM), utilizing Large Language Models (LLMs) and Text-to-Audio (TTA) models to represent genotypes as semantic descriptions [8]. This allows for the evolution of complex, realistic signals that go beyond traditional vector-based representations. Additionally, the role of sexual selection as a "magic trait" contributing to both ecological selection and non-random mating has been highlighted in the rapid evolution of prehistoric dogs from wolves [9]. This combination of evolutionary drivers explains how speciation can occur within relatively short timeframes, such as 15,000 years.

3 Problem Definition and Model Design

The core of this project is the development of a decentralized, Agent-Based Modeling (ABM) environment designed to simulate complex evolutionary dynamics. Unlike traditional evolutionary algorithms that operate on a global population level, this approach treats each individual as an autonomous entity with its own set of rules and spatial location. By shifting the focus to an agent-centered perspective, we can observe how macro-level phenomena, such as speciation and niche formation, emerge from simple, local-level behaviors and reproductive decisions.

To implement this simulation, the Mesa framework, a modular library for agent-based modeling in Python will be utilized. This framework provides the necessary infrastructure to manage a discrete-time simulation where agents inhabit a shared grid or continuous space, interact with their neighbors, and compete for resources or mating opportunities. The use of Mesa allows for a scalable and flexible architecture, making it possible to track the life cycle of thousands of agents and analyze the long-term stability of the population's diversity in a controlled, reproducible digital ecosystem.

3.1 System Architecture and Agent Specification

Building upon the Mesa framework, the simulation ecosystem is composed of two primary agent types: **Males** and **Females**. The biological and evolutionary state of each agent is defined by a specific set of attributes:

- **Traits:** Represented as a vector of values $T \in [0, 1]$. These define the phenotypic characteristics of the individual.
- **Preferences:** Exclusive to Female agents, these are represented by a preference vector of the same dimension as the trait vector ($P \in [0, 1]$), used as the primary mechanism for sexual selection.
- **Energy and Age:** Dynamic variables that determine the agent's viability. Energy is consumed through movement and reproduction, while age tracks the individual's progression through its life cycle.

The simulation environment is modeled as a **2D Continuous Toroidal Space** (1.0×1.0), as illustrated in Fig. 2. The toroidal topology ensures that agents can move across boundaries without encountering artificial edges, maintaining a seamless spatial flow.

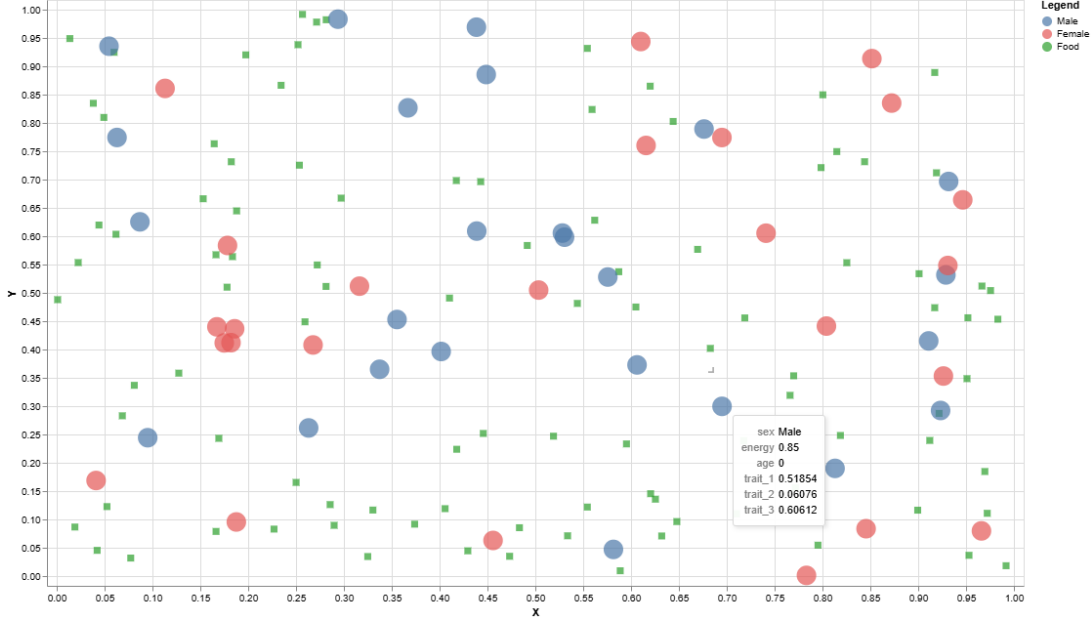


Figure 2: Visualization of the 2D toroidal environment containing agents (males and females) and food resources.

3.2 Behavioral Logic and Survival

The survival of the population depends on the availability of food resources, which regenerate at random locations during each iteration. Agents interact with these resources based on their internal physiological state:

1. **Foraging:** If an agent's energy drops below a critical threshold, it prioritizes survival by navigating toward the nearest food source to replenish its energy reserves. This metabolic necessity acts as a primary constraint on the agent's spatial behavior.
2. **Reproduction and Mate Choice:** When a Female agent accumulates sufficient energy and is not on cooldown, she initiates a search for a partner within her perception radius. The selection is based on **sexual selection**: she evaluates available Males and chooses the individual whose trait vector is closest to her own preference vector in terms of Euclidean distance.

$$d(T_m, P) = \|T_m - P\|_2 = \sqrt{\sum_{i=1}^d (T_{m,i} - P_i)^2}. \quad (1)$$

3. **Inheritance and Mutation:** Upon successful mating, a new agent is produced. The offspring's traits and preferences are not identical copies but are derived from the **arithmetic mean** of the parents' attributes, supplemented by a stochastic mutation factor:

$$T_{offspring} = \text{clip} \left(\frac{T_{female} + T_{male}}{2} + \mathcal{N}(0, \sigma^2), 0, 1 \right) \quad (2)$$

where σ represents the mutation standard deviation. This mechanism ensures that while offspring resemble their parents, new genetic variations are continuously introduced into the population, allowing for evolutionary shifts. The preference vector for female offspring is inherited directly from the mother, followed by a mutation step. This vertical transmission of preferences is what enables the co-evolutionary feedback loop.

4. **Mortality:** Population stability is governed by two death conditions. First, agents die if their energy reaches zero (*metabolic failure*). Second, the simulation incorporates a **senescence model** where the probability of death increases linearly with age, ensuring natural population turnover and preventing the indefinite survival of any single individual.
5. **Metabolic Trade-offs:** To simulate the biological "handicap principle," Male agents incur a continuous metabolic cost proportional to the magnitude of their trait vector:

$$E_{cost} = \alpha \cdot \|T_{male}\| \quad (3)$$

where α is the ornament cost coefficient. This ensures a trade-off between being attractive to Females and maintaining enough energy to survive, preventing the runaway selection from reaching extreme, non-viable values.

4 Experimental research results

This section details the experimental setup and analyzes the evolutionary dynamics observed during the simulations. To ensure statistical clarity and identify long-term trends, all results were processed using a moving average with a window size of 50 steps. This smoothing technique filters out stochastic noise inherent in agent-based modeling, allowing for a clearer interpretation of selection pressures and phenotypic shifts. All simulations were initialized with a balanced population size of 50 individuals.

4.1 Experiment 1: Coevolution of Traits and Preferences under Selection Pressure

The first experiment investigated the primary mechanism of sexual selection: how male ornamental traits align with female preferences over time when constrained by survival costs.

4.1.1 Experimental Setup

Key parameters included a moderate mutation rate ($\sigma = 0.05$) and a non-zero ornament cost ($\alpha = 0.03$), with a trait dimensionality of $d = 3$ and a simulation horizon of 2000 steps. The goal was to observe whether the "Fisherian Runaway" process would occur or if metabolic energy constraints would enforce phenotypic stabilization.

4.1.2 Results and Analysis

The analysis follows a hierarchy from global population stability to specific trait-dimensional interactions.

The immediate impact of selective pressure is visible in the population metrics (Fig. 3). As seen in Fig. 3a, the male genetic variance underwent a precipitous decline, dropping from 0.07 to approximately 0.01 within the first 250 steps. This elimination of high-variance traits indicates that only a narrow range of phenotypes was metabolically feasible under the $\alpha = 0.03$ cost constraint.

Furthermore, the *Trait-Preference Gap* (Fig. 3b) demonstrates that the population reached an "evolutionary stalemate." While the gap for Traits 2 and 3 minimized significantly, Trait 1 maintained a persistent offset. This suggests that for certain trait dimensions, the cost of

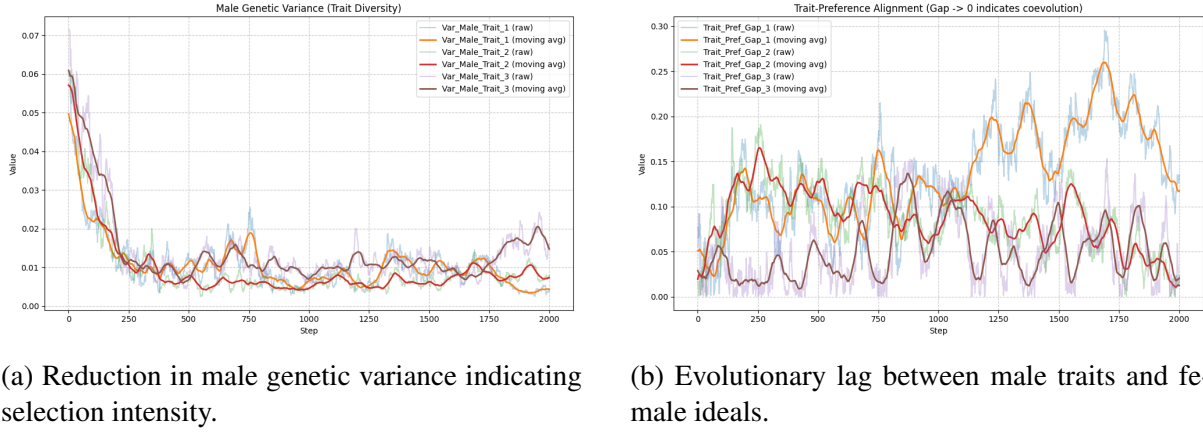


Figure 3: Experiment 1: Global metrics of population stabilization and selective pressure.

reaching the female ideal is higher than the reproductive benefit, resulting in a permanent evolutionary lag.

The conflict between sexual selection and natural selection is most evident when comparing individual trait dimensions (Fig. 4):

- **Trait 1 (Fig. 4a):** Exhibits a clear cost barrier. As female preferences trended upward, the male trait plateaued significantly below the preference line. This divergence shows that natural selection for energy conservation successfully counteracted the runaway sexual selection.
- **Trait 2 (Fig. 4b):** Demonstrates efficient tracking. The small residual gap between the preference and the trait indicates that, for this dimension, the marginal reproductive benefit of matching the preference outweighs the metabolic penalty.
- **Trait 3 (Fig. 4c):** Settles into a steady-state alignment, with both the male trait and the female preference fluctuating around a common equilibrium value. This is the closest analogue to a balanced Fisherian process observed within the experiment.

This experiment confirms that metabolic costs ($\alpha = 0.03$) act as a critical stabilizing force. The divergence observed in Trait 1 (Fig. 4a) illustrates that while sexual selection exerts a pull toward extremes, environmental constraints maintain the population within viable biological limits, preventing an uncontrolled runaway process.

4.2 Experiment 2: Survival Threshold Analysis

This experiment investigated the impact of increasing the ornament cost coefficient (α) on the demographic stability and long-term viability of the population. Simulations were conducted for $\alpha \in \{0.0, 0.025, 0.05, 0.075, 0.1\}$.

4.2.1 Experimental Setup

The initial population and environmental parameters remained consistent with previous runs to isolate the effect of the metabolic cost. The ornament cost was varied incrementally to identify the tipping point where sexual selection for high-value traits becomes a lethal burden. The primary metrics observed were the *sex ratio* and the *time to extinction*.

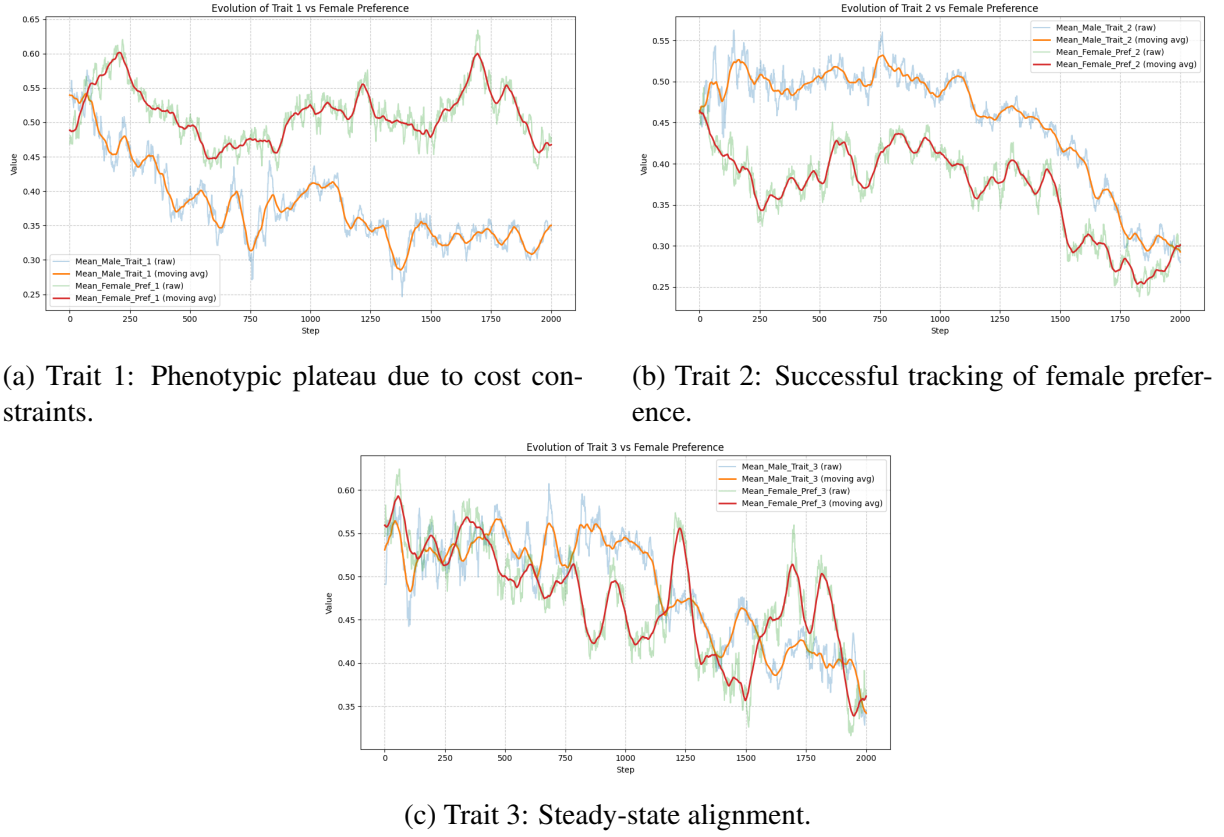


Figure 4: Experiment 1: Comparison of coevolutionary dynamics across three trait dimensions.

4.2.2 Results and Analysis

The data, summarized in Table 1 and visualized in Figure 5, reveals a clear transition from ecological stability to rapid population collapse as the cost of ornaments increases.

Cost (α)	Steps until Extinction	Population Status
0.000	2000+	Stable
0.025	2000+	Stable (High metabolic stress)
0.050	2000+	Significant sex disproportion
0.075	1074	Steady Extinction
0.100	628	Rapid Collapse

Table 1: Impact of ornament cost on population longevity.

The analysis indicates three distinct demographic zones:

- **Sustainability Zone** ($\alpha \leq 0.025$): As shown in Fig. 5a, the population maintains a healthy balance. While costs exist, males retain enough energy to ensure reproduction and survival.
- **Disproportion Zone** ($\alpha = 0.05$): The first signs of systemic failure appear. Although the population survives the 2000-step limit, a significant gap between the number of males and females develops. The energy required to maintain ornaments begins to reduce the average lifespan of males.
- **Collapse Zone** ($\alpha \geq 0.075$): The metabolic burden becomes unsustainable. In Fig. 5c, a death spiral is observed: males die faster than they are replaced by births, the dwindling

male pool starves females of mating opportunities, and the population count drops to zero (1074 steps for $\alpha = 0.075$ and only 628 steps for $\alpha = 0.1$).

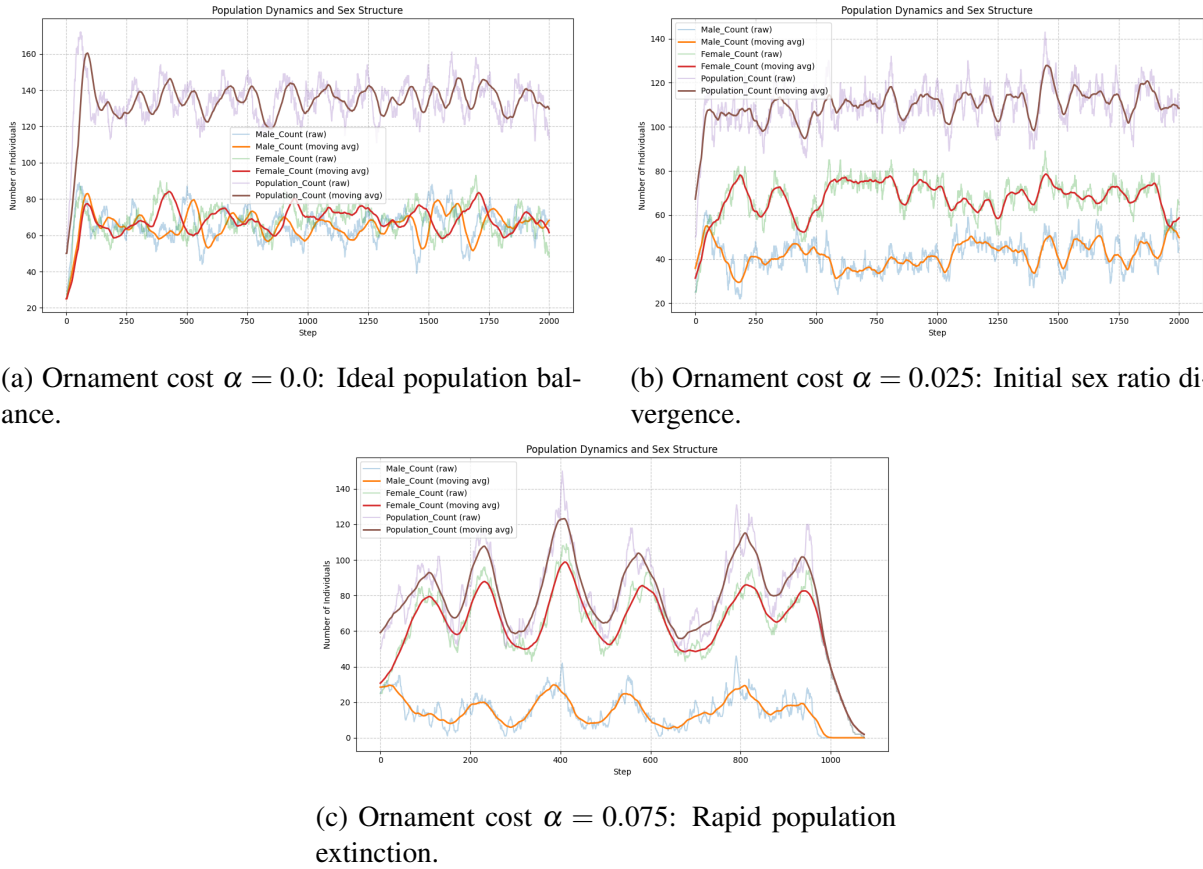


Figure 5: Experiment 2: Comparison of population dynamics under varying ornament cost levels.

The findings identify the Survival Threshold for this specific environment at approximately $\alpha \approx 0.05$. Beyond this value, the pressure of sexual selection forces males into a metabolic deficit that cannot be compensated by food consumption. This experiment demonstrates that sexual selection is not just a driver of trait evolution, but a potential factor in species extinction when environmental costs exceed a biological tipping point.

4.3 Experiment 3: Resource Abundance and the Cost of Reproduction

This experiment investigated how environmental resource levels affect the intensity of sexual selection and the physiological state of the population. Specifically, it tested the hypothesis that high resource availability leads to stronger selection at the cost of individual metabolic stability.

4.3.1 Experimental Setup

The simulation compared two distinct scenarios:

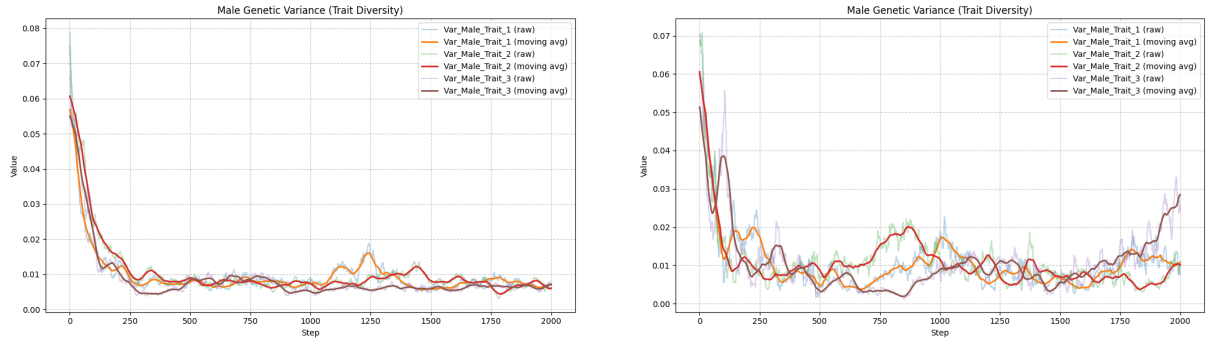
- **Wealth World:** High initial food count (50) and regrowth rate (15 food units per step), low movement cost (0.02) and a reduced female reproduction cost (0.5), creating a surplus of energy.

- **Desolate World:** Scarce initial food (20) and regrowth rate (8), elevated movement cost (0.04) and a higher female reproduction cost (0.7), forcing a survival-oriented strategy.

All genetic parameters, including initial trait values, mutation rate ($\sigma = 0.05$) and preference ranges, were kept identical, so that any divergence in the evolutionary outcome is attributable solely to the energetic regime of the environment.

4.3.2 Results and Analysis

The Wealth World (Fig. 6a) exhibits a rapid decline in male trait variance, as abundant food minimizes foraging constraints and allows females to exert strict mate preferences. This intense directional selection effectively purges genetic diversity. In contrast, the Desolate World (Fig. 6b) maintains high trait variance due to opportunistic mating. Here, the life-threatening search cost forces females to mate with the first available partner, resulting in "relaxed selection" that preserves a wider range of phenotypic traits.



(a) Wealth World: Low variance due to strict selection.

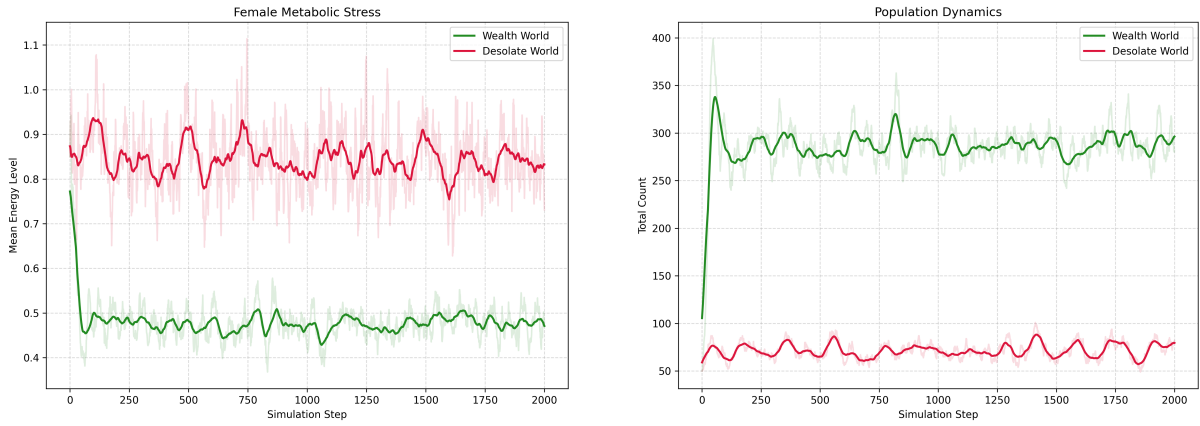
(b) Desolate World: High variance due to relaxed selection.

Figure 6: Experiment 3: Comparison of male trait variance between the Desolate and Wealth ecosystems.

A clear divergence in demographic stability is observed between the two ecosystems. The Wealth World reached a carrying capacity of approximately 282 individuals - roughly $3.5\times$ that of the Desolate World, which stabilized around 82 individuals (Fig. 7b). However, this demographic success directly correlates with a sustained depression in Mean Female Energy.

In the resource-rich environment, females engage in near-constant reproduction. Since each birth event imposes a significant metabolic tax (0.5 units), their energy reserves remain near the threshold of exhaustion (0.4–0.5 units). Conversely, females in the Desolate World maintain higher individual energy reserves (approx. 0.8 units) due to fewer mating opportunities, though this results in a stagnant and fragile population size.

Experiment 3 confirms that sexual selection is a luxury driven by environmental surplus. In the Desolate World, natural selection for survival overrides sexual preferences, maintaining genetic diversity but limiting growth. In the Wealth World, sexual selection is maximized, leading to rapid trait evolution and demographic expansion, but resulting in a population of metabolically stressed individuals who prioritize reproductive output over individual energy reserves.



(a) Metabolic Stress: Lower energy in Wealth World due to high birth rates.

(b) Population Dynamics: Rapid expansion vs. stagnation.

Figure 7: Experiment 3: Correlation between energy expenditure and population growth.

5 Summary and conclusions

The experiments demonstrate that the dynamics of sexually selected traits are governed by a complex three-way interaction between female preference strength, the metabolic cost of male ornamentation, and environmental resource availability. The first two studies validate the Handicap Principle: ornament costs effectively bound trait escalation, leading to a stable coevolutionary equilibrium rather than an unbounded runaway process. However, a critical survival threshold was identified at $\alpha \approx 0.05$, beyond which the metabolic burden on males triggers a demographic collapse. This suggests that while sexual selection is a powerful driver of phenotypic novelty, it can also function as a mechanism for self-inflicted extinction when the cost of attractiveness exceeds the system's energetic capacity.

The third experiment highlights that the intensity of sexual selection is fundamentally gated by ecological surplus. In the Wealth World, abundant resources minimized search costs, enabling strict mate choice that resulted in rapid directional evolution and population expansion, albeit at the cost of chronically depressed female energy reserves due to constant reproduction. Conversely, the Desolate World forced a shift toward reproductive opportunism and relaxed selection, preserving genetic diversity but limiting population growth. These findings position sexual selection as a parameter-tunable mechanism for maintaining diversity in evolutionary computation, provided that the energetic boundary conditions of the environment are maintained above a critical viability threshold.

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